

Tianze Yin

548-883-1207 | yintz1207@gmail.com | [linkedin.com/in/tianze-yin](https://www.linkedin.com/in/tianze-yin) | github.com/tianzeyin

SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL, HTML/CSS, Verilog

Frameworks & Libraries: Next.js, React, FastAPI, Playwright, Better Auth, SQLModel

Infrastructure & Tools: PostgreSQL, CDN, Object Storage, Nginx, PM2, FFmpeg, HLS

WORK EXPERIENCE

Software Engineer Intern [*Next.js, PostgreSQL, Better Auth, CDN, HLS, Nginx*] May 2026 – Present
YGBK International Education Technology ￼ *Beijing, China*

- Built and deployed a **production full-stack education and digital library platform** serving **10,000+ users** across the web and **WeChat Mini Program**.
- Developed **100+ API route handlers** and **50+ PostgreSQL migrations** supporting books, courses, classes, user management, learning progress, reporting, and administrative workflows.
- Engineered **secure audiobook streaming** using **FFmpeg and encrypted HLS**, with segmented delivery, expiring playback sessions, protected key access, and private object storage.
- Integrated a **CDN** with signed URLs and edge caching, and deployed the platform using **Nginx and PM2** with automated deployments, health checks, logging, background workers, and rollback protection.

QA Engineer [*React, Next.js, Playwright, TypeScript, Python*] Sept 2024 – Jun 2025
Kortex ￼ *Toronto, Ontario*

- Automated **end-to-end tests** for complex frontend interactions using **Playwright**, identifying, reproducing, and documenting defects across application features.
- Expanded **automated test coverage** for core user workflows, including document creation and deletion, billing interactions, context menus, and bulk actions.
- Tested **React and Next.js** interfaces by isolating frontend issues and validating fixes, edge cases, and regression stability through automated **Playwright test suites**.

Software Developer [*FastAPI, Next.js, Python, SQLModel*] Dec 2025 – Present
University of Waterloo Orbital Design Team ￼ *Waterloo, Ontario*

- Developed backend command-management features for the **Ground Station team**, building **FastAPI endpoints** to create, view, delete, and manage spacecraft commands.
- Implemented **command parameter and format validation** and used **SQLModel to persist command data**, ensuring only validated commands were stored consistently in the database.
- Added **command status tracking, timestamps, and request logging** to improve traceability, backend monitoring, and debugging across spacecraft command workflows.

PROJECTS

True Light ￼ | *React, Expo, FastAPI, OpenCV, TypeScript, Python* Jan 2026

- Built a **real time mobile accessibility app** at **DeltaHacks** in **72 hours**, provided visual and audio feedback to help colorblind and low-vision users identify colors in their surroundings.
- Developed an **Expo React frontend** with live camera capture and **OpenCV contour and color analysis** to detect and classify more than **30 color regions** from real-time image input.
- Integrated a **FastAPI image processing backend** with **ElevenLabs text-to-speech** to transform detected visual information into immediate spoken accessibility feedback for users.

Sprout ￼ | *React, Next.js, Tailwind CSS, ElevenLabs, Claude API* Feb 2026

- Placed **3rd** in the **University of Waterloo AI & Data Science Hackathon** by building a English and Rohingya learning platform for newcomers and refugees with low literacy.
- Developed core learning features including **speaking assessment**, conversation practice, **AI assisted vocabulary generation**, and user progress tracking.
- Integrated **ElevenLabs** and the **Claude API** to support live voice conversations and deliver **adaptive AI generated learning feedback** based on each user's spoken input and practice interactions.

EDUCATION

University of Waterloo, BAsC in Computer Engineering

Sept 2025 – May 2030